Computer Vision and Image Processing

Mukesh A. Zaveri
Computer Engineering Department
Sardar Vallabhbhai National Institute of Technology, Surat
mazaveri@coed.svnit.ac.in



Image Formation



Image Brightness

- image of 3D object
  - shape
  - reflectance properties
  - distribution of light sources
  - position of object relative to the imaging system
- position and distribution of the source of illumination determines brightness pattern
- need to know how image is formed
  - brightness at a particular point in the image
- required to know Radiometry

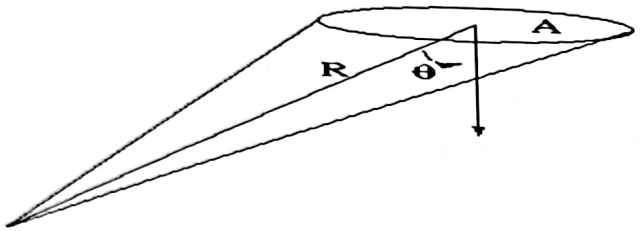


Radiometry: Irradiance and Radiance

- irradiance: amount of light falling on a surface
  - power per unit area incident on the surface (watts per square meter)
- radiance: amount of light radiated from a surface
  - power per unit area per unit solid angle emitted from the surface (watts per square meter per steradian)
  - surface can radiate into a whole hemisphere of possible directions and can radiate different amount of energy in different directions
  - solid angle of a cone of directions is defined as the area cut out by the cone on the unit sphere
  - a hemisphere of directions has a solid angle 2π



Radiometry: Solid Angle



- A small planar patch of area A at distance R from the origin subtends a solid angle

$$\Omega = \frac{A\cos\theta}{R^2}$$

θ is the angle between a surface normal and a line connecting the patch to the origin

- foreshortened area - the surface area multiplied by the cosine of the angle between a perpendicular to the surface and the specified direction

Image Formation

- the solid angle of the cone of rays leading to the patch on the object is equal to the solid angle of the cone of rays leading to image patch
- apparent area of the image patch as seen from the center of the lens is $\delta I \cos\alpha$
- the distance from center of the lens is $f/\cos\alpha$
- solid angle subtended by the image patch

$$\frac{\delta I\cos\alpha}{(f/\cos\alpha)^2}$$

- the solid angle of the object patch as seen from the center of the lens is

$$\frac{\delta O\cos\theta}{(z/\cos\alpha)^2}$$

- with two solid angles are to be equal

$$\frac{\delta O}{\delta I} = \frac{\cos\alpha}{\cos\theta}\left(\frac{z}{f}\right)^2$$

Image Formation

- Brightness is determined by the amount of energy an imaging system receives per unit apparent area
- relation between the radiance at a point on an object (scene radiance) and the irradiance at the corresponding point in the image (image irradiance)

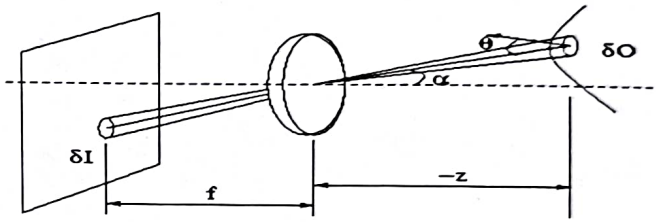
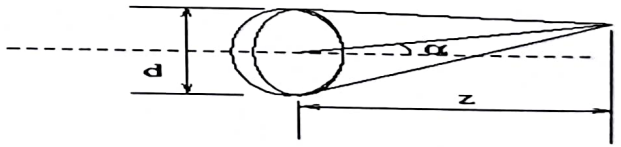


Image Formation

- how much of the light emitted by the surface makes its way through the lens



- solid angle subtended by the lens as seen from the object patch

$$\Omega = \frac{\pi}{4}\frac{d^2\cos\alpha}{(z/\cos\alpha)^2} = \frac{\pi}{4}\left(\frac{d}{z}\right)^2\cos^3\alpha$$

- the power of the light originating on the patch and passing through the lens is

$$\delta P = L\delta O\Omega\cos\theta = L\delta O\frac{\pi}{4}\left(\frac{d}{z}\right)^2\cos^3\alpha\cos\theta$$

L is the radiance of the surface in the direction toward the lens. This power is concentrated in the image.

Image Formation

- no light from other areas reaches this image patch

$$E = \frac{\delta P}{\delta I} = L\frac{\delta O}{\delta I}\frac{\pi}{4}\left(\frac{d}{z}\right)^2 \cos^3\alpha\cos\theta$$

E is the irradiance of the image at the patch, substituting $\delta O/\delta I$

$$E = L\frac{\pi}{4}\left(\frac{d}{f}\right)^2 \cos^4\alpha$$

- image irradiance is proportional to scene radiance
- this relationship exploited to recover information about object from its image
- inverse of the square of the effective f-number (f/d)
- brightness - image irradiance E is proportional to scene radiance L

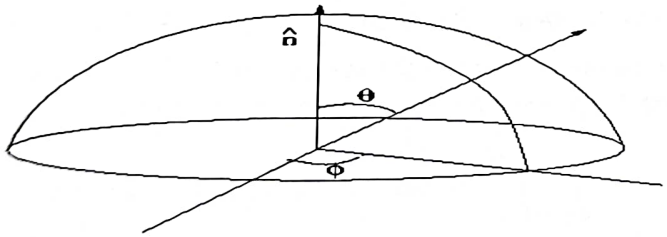


Bidirectional Reflectance Distribution Function (BRDF)

- scene radiance depends on
  - the amount of light that falls on a surface and the fraction of the incident light that is reflected
  - geometry of light reflection
- In short, radiance of a surface depends on the direction from which it is viewed and the direction from which it is illuminated
- surface normal $\hat{n}$, θ between a ray and the normal and ϕ between a perpendicular projection of the ray onto the surface and the reference line on the surface
- θ - polar angle
- ϕ - azimuth angle



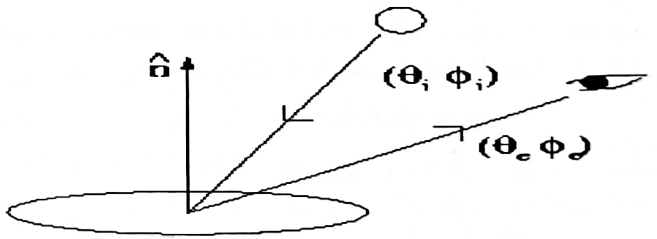
BRDF



- the direction (θ_i, ϕ_i) from which light is falling on the surface
- the direction (θ_e, ϕ_e) into which it is emitted toward the viewer



BRDF



- BRDF $f(\theta_i, \phi_i, \theta_e, \phi_e)$ tells us how bright a surface appears when viewed from one direction while light falls on it from another
- the amount of light falling on the surface from the direction (θ_i, ϕ_i) the irradiance be $\delta E(\theta_i, \phi_i)$
- the brightness of the surface as seen from the direction (θ_e, ϕ_e) the radiance be $\delta L(\theta_e, \phi_e)$
- BRDF is the ratio of radiance to irradiance

$$f(\theta_i, \phi_i, \theta_e, \phi_e) = \frac{\delta L(\theta_e, \phi_e)}{\delta E(\theta_i, \phi_i)}$$



BRDF: Surfaces

- for many surfaces the radiance is not altered if the surface is rotated about the surface normal
- BRDF depends only on the difference $\phi_e - \phi_i$
- this is true for matte surfaces and specularly reflecting surfaces but not for surfaces with oriented microstructure (tiger's eye and iridescent feathers of some birds)
- constraint on BRDF: if two surfaces are in thermal equilibrium, radiation reaching one from the other must be balanced by radiation flowing in the opposite direction
- otherwise surface receiving more radiation heat up and other would cool down which disturbs the equilibrium (second law of thermodynamics)
- BRDF is constrained by Helmholtz reciprocity condition

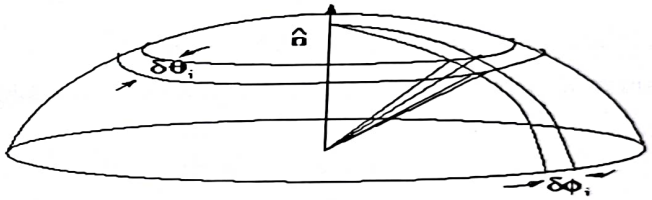
$$f(\theta_i, \phi_i; \theta_e, \phi_e) = f(\theta_e, \phi_e; \theta_i, \phi_i)$$



Extended Light Sources

- previously assumed that all light comes from one direction
- several light sources or extended source (sky)
- nonzero solid angle of directions required to obtain a nonzero radiance
- infinitesimal patch of the sky, of size $\delta\theta_i$ in polar angle and $\delta\phi_i$ in azimuth and this patch subtends a solid angle

$$\delta\omega = \sin\theta_i \delta\theta_i \delta\phi_i$$





BRDF: Extended Light Sources

- $E(\theta_i, \phi_i)$ be the radiance per unit solid angle coming from the direction (θ_i, ϕ_i), then the radiance from the patch is

$$E(\theta_i, \phi_i) \sin\theta_i \delta\theta_i \delta\phi_i$$

- the total irradiance of the surface is

$$E_0 = \int_{-\pi}^{\pi} \int_{0}^{\pi/2} E(\theta_i, \phi_i) \sin\theta_i \cos\theta_i d\theta_i d\phi_i$$

$\cos\theta_i$ term accounts for the foreshortening of the surface as seen from the direction (θ_i, ϕ_i)

- for radiance - integrate the product of the BRDF and the irradiance over the hemisphere of possible directions from which light can fall on a surface

$$L(\theta_e, \phi_e) = \int_{-\pi}^{\pi} \int_{0}^{\pi/2} f(\theta_i, \phi_i; \theta_e, \phi_e) E(\theta_i, \phi_i) \sin\theta_i \cos\theta_i d\theta_i d\phi_i$$

function of two variable θ_e and ϕ_e; specify the direction of the ray emitted toward the viewer



BRDF: Extended Light Sources

- ideal Lambertian surface is one that appears equally bright from all viewing directions and reflects all incident light, absorbing none, i.e., BRDF must be a constant for such a surface
- integrate the radiance of the surface over all directions and equate the total radiance so obtained to the total irradiance

$$\int_{-\pi}^{\pi} \int_{0}^{\pi/2} fE \cos\theta_i \sin\theta_e \cos\theta_e d\theta_e d\phi_e = E \cos\theta_i$$

or

$$2\pi f \int_{0}^{\pi/2} \sin\theta_e \cos\theta_e d\theta_e d\phi_e = 1$$

- $\pi f = 1$ for an ideal Lambertian surface, BRDF is constant $f(\theta_i, \phi_i; \theta_e, \phi_e) = \frac{1}{\pi}$
- the radiance $L = \frac{1}{\pi} E_0$



Reflectance Map

- makes explicit relationship between surface orientation and brightness
- encodes information about surface reflectance properties and light-source distributions
- representational tool for recovering surface shape from images
- a source of radiance E illuminating a Lambertian surface
- scened radiance $L = \frac{1}{\pi} E \cos\theta_i$ (θ_i is angle between the surface normal and the direction toward source)

$$\cos\theta_i = \frac{1 + p_s p + q_s q}{\sqrt{1 + p^2 + q^2}\sqrt{1 + p_s^2 + q_s^2}}$$

- called reflectance map $R(p, q)$; depends on the properties of surface material of object and the distribution of light sources



Reflectance Map

- Image irradiance is proportional to a number of constants; the inverse of the square of f-number and the fixed brightness of the source
- for this reason, the reflectance map is normalized in some way, having maximum value one
- for Lambertian surface illuminated by a single distant point source $R(p, q)$ is defined as above
- reflectance map gives the dependence of scene radiance on surface orientation
- plot surface $R(p, q)$ - function of gradient (p, q); pq plane called gradient space
- every point in it corresponds to a particular surface orientation
- the point at the origin represents the orientation of all planes that are perpendicular to the viewing direction



Reflectance Map

- contour map in gradient space used to depict a reflectance map
- Lambertian surface; contours of constant brightness are nested conic sections in pq-plane $R(p, q) = c$

$$(1 + p_s p + q_s q)^2 = c^2(1 + p^2 + q^2)(1 + p_s^2 + q_s^2)$$

- $R(p, q)$ is maximum at $(p, q) = (p_s, q_s)$
- Shading in Images
  - brightness pattern in image affected by shape of an object
  - surface patches with different orientations appear with different brightness
  - the variation of brightness is called shading
  - reflectance map captures the dependence of brightness on surface orientation



Reflectance Map

- image irradiance $E(x, y)$ a particular point in the image proportional to the radiance at the corresponding point on the surface imaged
- surface gradient at that point is (p, q); the radiance there is $R(p, q)$
- normalizing by setting the constant of proportionality to one

$$E(x, y) = R(x, y)$$

- sphere with a Lambertian surface illuminated by a point source at essentially the same place as the viewer
- $\theta_e = \theta_i$ and $(p_s, q_s) = (0, 0)$

$$R(p, q) = \frac{1}{\sqrt{1 + p^2 + q^2}}$$



Reflectance Map: Example

- with sphere on the optical axis; equation for its surface

$$z = z_0 + \sqrt{r^2 - (x^2 + y^2)}$$

r is radius, z_0 is the distance of its center from the lens and $x^2 + y^2 \leq r^2$

$$p = -\frac{x}{z - z_0} \quad \text{and} \quad q = -\frac{y}{z - z_0}$$

$$E(x, y) = R(p, q) = \frac{z - z_0}{r} = \sqrt{1 - \frac{x^2 + y^2}{r^2}}$$

- sphere with different reflectance properties, say,

$$R(p, q) = \frac{1 + p_s p + q_s q}{\sqrt{1 + p_s^2 + q_s^2}}$$

$(p_s, q_s) = (0, 0)$ obtains a uniformly bright disk in the image



Photometric Stereo

- reflectance map useful in CG: image is created from a description of the shape of an object
- Given an image recover the shape
- unique mapping from surface orientation specified by (p, q) to radiance, given by the reflectance map $R(p, q)$
- but inverse mapping is not unique; an infinite number of surface orientations give rise to the same brightness
- contour of constant $R(p, q)$ connects such a set of orientations in the reflectance map
- surface orientation can be determined uniquely for some special points; where brightness is maximum or minimum of $R(p, q)$ Lambertian surface $R(p, q) = 1$ only when $(p, q) = (p_s, q_s)$
- mapping from brightness to surface orientation cannot be unique, since brightness has one degree of freedom while orientation has two.



Photometric Stereo

- to determine two unknowns p and q; need two equations
- two images with different lighting yield two equations for each image point

$$R_1(p, q) = E_1 \quad \text{and} \quad R_2(p, q) = E_2$$

- these equations are linear and independent then unique solution for p and q is possible

$$R_1(p, q) = \sqrt{\frac{1 + p_1 p + q_1 q}{r_1}} \quad \text{and} \quad R_2(p, q) = \sqrt{\frac{1 + p_2 p + q_2 q}{r_2}}$$

$$r_1 = \sqrt{1 + p_1^2 + q_1^2} \quad \text{and} \quad r_2 = \sqrt{1 + p_2^2 + q_2^2}$$

$$p = \frac{(E_1^2 r_1 - 1)q_2 - (E_2^2 r_2 - 1)q_1}{p_1 q_2 - q_1 p_2} \quad q = \frac{(E_2^2 r_2 - 1)p_1 - (E_1^2 r_1 - 1)p_2}{p_1 q_2 - q_1 p_2}$$



Photometric Stereo

- if equations are nonlinear; no solutions or several solutions

$$R_1(p, q) = \frac{1 + p_1 p + q_1 q}{\sqrt{1 + p^2 + q^2}\sqrt{1 + p_1^2 + q_1^2}}$$

$$R_2(p, q) = \frac{1 + p_2 p + q_2 q}{\sqrt{1 + p^2 + q^2}\sqrt{1 + p_2^2 + q_2^2}}$$

- can be two solutions, one solution or none depending on the values of R_1 and R_2



Recovering Albedo

- surface with non-uniform reflectance properties
- radiance is product of reflectance factor (albedo) and some function of orientation
- Albedo: a number between zero and one that indicates how much light the surface reflects relative to some ideal surface with the same geometric dependence in the BRDF
- matte surface like Lambertian surface but does not reflect all of the incident light
- its brightness is $\rho \cos\theta_i$
- recover ρ and (p, q) need three images



Recovering Albedo

- unit vectors in the directions of three source positions

$$\hat{\mathbf{s}}_i = \frac{(-p_i, -q_i, 1)^T}{\sqrt{1 + p_i^2 + q_i^2}} \qquad i = 1, 2, 3$$

$$E_i = \rho(\hat{\mathbf{s}}_i \cdot \hat{\mathbf{n}}) \quad \hat{\mathbf{n}} = \frac{(-p, -q, 1)^T}{\sqrt{1 + p^2 + q^2}}$$

$$E_1 = \rho(\hat{\mathbf{s}}_1 \cdot \hat{\mathbf{n}}) \;\; E_2 = \rho(\hat{\mathbf{s}}_2 \cdot \hat{\mathbf{n}}) \;\; E_3 = \rho(\hat{\mathbf{s}}_3 \cdot \hat{\mathbf{n}})$$

$$\mathbf{E} = \rho \mathbf{S} \hat{\mathbf{n}} \quad \rho \hat{\mathbf{n}} = \mathbf{S}^{-1} \mathbf{E}$$

$$\rho \hat{\mathbf{n}} = \frac{1}{[\hat{\mathbf{s}}_1 \hat{\mathbf{s}}_2 \hat{\mathbf{s}}_3]} (E_1(\hat{\mathbf{s}}_2 \times \hat{\mathbf{s}}_3) + E_2(\hat{\mathbf{s}}_3 \times \hat{\mathbf{s}}_1) + E_3(\hat{\mathbf{s}}_1 \times \hat{\mathbf{s}}_2))$$

where $[\hat{\mathbf{s}}_1 \hat{\mathbf{s}}_2 \hat{\mathbf{s}}_3]$ is the triple product $\hat{\mathbf{s}}_1 \cdot (\hat{\mathbf{s}}_2 \times \hat{\mathbf{s}}_3)$

- unique result is assured



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Optical Flow

- The apparent motion of brightness patterns observed when a camera is moving relative to the objects being images is called the optical flow
- when object move in front of a camera or a camera moves through a fixed environment; corresponding changes in the image
- these changes used to recover the relative motion and the shape of the object
- motion field - assign a velocity vector to each point in the image
- object point P_o have velocity $\mathbf{v}_o$ relative to the camera; point moves $\mathbf{v}_o \delta t$
- induces a motion $\mathbf{v}_i$ in the corresponding image point P_i; it moves $\mathbf{v}_i \delta t$

$$\mathbf{v}_o = \frac{d\mathbf{r}_o}{dt} \text{ and } \mathbf{v}_i = \frac{d\mathbf{r}_i}{dt}$$

- $\mathbf{r}_o$ and $\mathbf{r}_i$ related by

$$\frac{1}{f'}\mathbf{r}_i = \frac{1}{\mathbf{r}_o \cdot \hat{\mathbf{z}}}\mathbf{r}_o$$



Optical Flow



Optical Flow

- differentiating of this perspective projection

$$\frac{1}{f'}\mathbf{v}_i = \frac{(\mathbf{r}_o \cdot \hat{\mathbf{z}})\mathbf{v}_o - (\mathbf{v}_o \cdot \hat{\mathbf{z}})\mathbf{r}_o}{(\mathbf{r}_o \cdot \hat{\mathbf{z}})^2} = \frac{(\mathbf{r}_o \times \mathbf{v}_o) \times \hat{\mathbf{z}}}{(\mathbf{r}_o \cdot \hat{\mathbf{z}})^2}$$

- these vectors constitute the motion field
- neighboring points on an object have similar velocities; so induced motion field in the image is also continuous in most places
- brightness patterns moves as object that give rise to them move
- ideally the optical flow will correspond to the motion field
- e.g. uniform sphere, a patch of uniform brightness - optical flow is zero everywhere despite a nonzero motion field
- $E(x, y, t)$ irradiance at image point (x, y) at t, $u(x, y)$ and $v(x, y)$ are the x and y components of the optical flow
- at $t + \delta t$ the point is at $(x + \delta x, y + \delta y)$; $\delta x = u\delta t$ and $\delta y = v\delta t$



Optical Flow

$$E(x+u\delta t, y+v\delta t, t+\delta t) = E(x,y,t)$$

- expand the left-hand side of equation in a Taylor series

$$E(x,y,t) + \delta x\frac{\partial E}{\partial x} + \delta y\frac{\partial E}{\partial y} + \delta t\frac{\partial E}{\partial t} + e = E(x,y,t)$$

- e represents high-order terms, dividing by δt and taking the limit as $\delta t \to 0$

$$\frac{\partial E}{\partial x}\frac{dx}{dt} + \frac{\partial E}{\partial y}\frac{dy}{dt} + \frac{\partial E}{\partial t} = 0 \quad \frac{dE}{dt} = 0$$

$$u = \frac{dx}{dt} \quad v = \frac{dy}{dt} \quad E_x = \frac{\partial E}{\partial x} \quad E_y = \frac{\partial E}{\partial y} \quad E_t = \frac{\partial E}{\partial t}$$

$$E_x u + E_y v + E_t = 0$$

- E_x, E_y and E_t are estimated from the image and the equation is called the optical flow constraint equation
- constraint on the components u and v of the optical flow



Optical Flow: Constraints

- additional constraint - rigid body motion is very restrictive and provides a powerful constraint
- motion field varies smoothly in most parts of the image
- minimize a measure of departure from smoothness

$$e_s = \int\int((u_x^2 + u_y^2) + (v_x^2 + v_y^2))dxdy$$

the integral of the square of the magnitude of the gradient of the optical flow

- error in the optical flow constraint equation should also be small

$$e_c = \int\int(E_x u + E_y v + E_t)^2 dxdy$$

- minimize $e_s + \lambda e_c$ λ weights the error in the image motion equation relative to the departure from smoothness
- it is large if brightness measurements are accurate and small if they are noisy



Aperture problem

- velocity space; values (u, v) satisfying the constraint equation lie on a straight line in velocity space

$$(E_x, E_y) \cdot (u, v) = -E_t$$

- component of optical flow in the direction of the brightness gradient $(E_x, E_y)^T$ is

$$\frac{E_t}{\sqrt{E_x^2 + E_y^2}}$$

- component of the optical flow at right angle to this direction can not be determined, that is, along the isobrightness contour
- this ambiguity is known as Aperture problem



Optical Flow: Solution

- minimizing an integral of the form

$$\int\int F(u, v, u_x, u_y, v_x, v_y)dxdy$$

is a problem in the calculus of variation and corresponding Euler equations are

$$F_u - \frac{\partial}{\partial x}F_{u_x} - \frac{\partial}{\partial y}F_{u_y} = 0 \quad F_v - \frac{\partial}{\partial y}F_{v_x} - \frac{\partial}{\partial y}F_{v_y} = 0$$

- In our case,

$$F = (u_x^2 + u_y^2) + (v_x^2 + v_y^2) + \lambda(E_x u + E_y v + E_t)^2$$

yield following equations and can be solved using iterative methods

$$\nabla^2 u = \lambda(E_x u + E_y v + E_t)E_x \quad \nabla^2 v = \lambda(E_x u + E_y v + E_t)E_y$$

$(\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2})$ is Laplacian operator



Discrete Case

- using the values of the optical flow at the grid point (i,j) and its neighbors allows us to measure the departure from smoothness as

$$s_{i,j} = \frac{1}{4}((u_{i+1,j}-u_{i,j})^2+(u_{i,j+1}-u_{i,j})^2+(v_{i+1,j}-v_{i,j})^2+(v_{i+1,j}-v_{i,j})^2)$$

- error in the optical flow constraint equation is

$$c_{ij} = (E_x u_{ij} + E_y v_{ij} + E_t)^2$$

E_x, E_y and E_t are estimates of the rates of change of brightness with respect to x, y and t at the point (i,j)

- seek a set of values $\{u_{ij}\}$ and $\{v_{ij}\}$ that minimize

$$e\sum_i\sum_j(s_{ij}+\lambda c_{ij})$$



Discrete Case

- differentiating e with respect to u_{kl} and v_{kl}

$$\frac{\partial e}{\partial u_{kl}} = 2(u_{kl}-\bar{u}_{kl}) + 2\lambda(E_x u_{kl} + E_y v_{kl} + E_t)E_x$$

$$\frac{\partial e}{\partial v_{kl}} = 2(v_{kl}-\bar{v}_{kl}) + 2\lambda(E_x u_{kl} + E_y v_{kl} + E_t)E_y$$

$\bar{u}$ and $\bar{v}$ are local averages

- the extremum occurs where the above derivatives of e are zero

$$(1+\lambda E_x^2)u_{kl} + \lambda E_x E_y v_{kl} = \bar{u}_{kl} - \lambda E_x E_t$$

$$\lambda E_y E_x u_{kl} + (1+\lambda E_y^2)v_{kl} = \bar{v}_{kl} - \lambda E_y E_t$$

- pair of equations in u_{kl} and v_{kl}; the determinant of the 2×2 coefficient matrix is

$$1+\lambda(E_x^2+E_y^2)$$

so that



Discrete Case: Solution

$$(1+\lambda(E_x^2+E_y^2))u_{kl} = +(1+\lambda E_y^2)\bar{u}_{kl} - \lambda E_x E_y \bar{v}_{kl} - \lambda E_x E_t$$

$$(1+\lambda(E_x^2+E_y^2))v_{kl} = -\lambda E_y E_x \bar{u}_{kl} + (1+\lambda E_x^2)\bar{v}_{kl} - \lambda E_y E_t$$

solve these equations for u_{kl} and v_{kl}

- iterative scheme

$$u_{kl}^{n+1} = \bar{u}_{kl}^n - \frac{E_x\bar{u}_{kl}^n + E_y\bar{v}_{kl}^n + E_t}{1+\lambda(E_x^2+E_y^2)}E_x \quad v_{kl}^{n+1} = \bar{v}_{kl}^n - \frac{E_x\bar{u}_{kl}^n + E_y\bar{v}_{kl}^n + E_t}{1+\lambda(E_x^2+E_y^2)}E_y$$

- new value of (u,v) at a point is set equal to the averages of the surrounding values, minus an adjustment, which in velocity space is in the direction of the brightness gradient
- consistent estimates of spatial and time derivatives of brightness

$$\begin{aligned} E_x \approx\ & \frac{1}{4\delta x}(E_{i+1,j,k} + E_{i+1,j,k+1} + E_{i+1,j+1,k} + E_{i+1,j+1,k+1}) \\ & -\frac{1}{4\delta x}(E_{i,j,k} + E_{i,j,k+1} + E_{i,j+1,k} + E_{i,j+1,k+1}) \end{aligned}$$



Discrete Case: Solution

$$\begin{aligned} E_y \approx\ & \frac{1}{4\delta y}(E_{i,j+1,k} + E_{i,j+1,k+1} + E_{i+1,j+1,k} + E_{i+1,j+1,k+1}) \\ & -\frac{1}{4\delta y}(E_{i,j,k} + E_{i,j,k+1} + E_{i+1,j,k} + E_{i+1,j,k+1}) \end{aligned}$$

$$\begin{aligned} E_t \approx\ & \frac{1}{4\delta t}(E_{i,j,k+1} + E_{i,j+1,k+1} + E_{i+1,j,k+1} + E_{i+1,j+1,k+1}) \\ & -\frac{1}{4\delta t}(E_{i,j,k} + E_{i,j+1,k} + E_{i+1,j,k} + E_{i+1,j+1,k}) \end{aligned}$$

- the optical flow velocities will be estimated at points lying between picture cells and between successive frames



Computer Vision and Image Processing

Mukesh A. Zaveri
Computer Engineering Department
Sardar Vallabhbhai National Institute of Technology, Surat
mazaveri@coed.svnit.ac.in



Shape from Shading

- recovery of surface shape from a single image
- linear reflectance map and reflectance map rotationally symmetric
- image irradiance equation - nonlinear first order partial differential equation
- traditional method using growing characteristic strips, sequential process
- for parallel algorithm - formulate a minimization problem that leads to a relaxation algorithm on a grid
  - minimize the integral of the difference between the observed brightness and that predicted for the estimated shape



Shape from Shading



Rotationally Symmetric Reflectance Maps

- light source is distributed in a rotationally symmetric fashion about the viewer
- e.g. hemispherical sky; viewer is looking straight down from above, point source at essentially the same place as the viewer

$$R(p,q) = f(p^2+q^2)$$

- f is monotonic and differentiable, with inverse f^{-1}

$$p^2+q^2 = f^{-1}(E(x,y))$$

- the direction of steepest ascent makes an angle θ_s with the x-axis where $\tan\theta_s = q/p$

$$\cos\theta_s = p/\sqrt{p^2+q^2} \text{ and } \sin\theta_s = q/\sqrt{p^2+q^2}$$

- slope of the surface in the direction of steepest ascent

$$m(\theta_s) = \sqrt{p^2+q^2} = \sqrt{f^{-1}(E(x,y))}$$



Reflectance Map

- given brightness we can find the slope of the surface but not the direction of steepest ascent
- if the direction of steepest ascent, given by (p, q) is known
- a small step of length $\delta\xi$ in the direction of steepest ascent; the changes in x and y

$$\delta x = p/\sqrt{p^2+q^2}\delta\xi \text{ and } \delta y = q/\sqrt{p^2+q^2}\delta\xi$$

- the changes in z

$$\delta z = m\delta\xi = \sqrt{p^2+q^2}\delta\xi = \sqrt{f^{-1}(E(x,y))}\delta\xi$$

- simplify by taking a step of length $\sqrt{p^2+q^2}\delta\xi$ rather than $\delta\xi$

$$\delta x = p\delta\xi \;\; \delta y = q\delta\xi \;\; \delta z = (p^2+q^2)\delta\xi = f^{-1}(E(x,y))\delta\xi$$

- problem: need to determine the values of p and q at the new point in order to continue the solution
- need to develop equations for the changes δp and δq in p and q



Characteristic Curves

- substituting δx and δy

$$\delta p = (pr+qs)\delta\xi = \frac{E_x}{2f'}\delta\xi \text{ and } \delta q = (ps+qt)\delta\xi = \frac{E_y}{2f'}\delta\xi$$

- as $\delta\xi \to 0$, obtain differential equations

$$\dot{x} = p \;\; \dot{y} = q \;\; \dot{z} = (p^2+q^2) \;\; \dot{p} = \frac{E_x}{2f'} \;\; \dot{q} = \frac{E_y}{2f'}$$

dot denotes differentiation with respect to ξ

- Given starting values, these five differential equations can be solved numerically to produce a curve on the surface of the object
- these curves are called characteristic curves (in this case they happen to be the curves of steepest ascent)

$$\ddot{x} = \frac{E_x}{2f'} \;\; \ddot{y} = \frac{E_y}{2f'} \;\; \dot{z} = f^{-1}(E(x,y))$$

these equations can be solved numerically



Brightness Gradient

- planar surface patch gives rise to a region of uniform brightness in the image
- nonzero brightness gradient $(E_x, E_y)^T$ occurs only where the surface is curved
- for brightness gradient, differentiate the image irradiance equation with respect to x and y

$$E(x,y) = f(p^2+q^2)$$

$$r = \frac{\partial^2 z}{\partial x^2} \;\; s = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} \;\; t = \frac{\partial^2 z}{\partial y^2}$$

- using the chain rule for differentiation

$$E_x = 2(pr+qs)f' \text{ and } E_y = 2(ps+qt)f'$$

- to determine the changes δp and δq occasioned by the step $(\delta x, \delta y)$ in the image plane

$$\delta p = r\delta x + s\delta y \text{ and } \delta q = s\delta x + t\delta y$$



Recovering Depth

- In general, knowing the coordinates of a particular point on the surface the solution can be extended; by taking a small step $(\delta x, \delta y)$, the change in depth

$$\delta z = p\delta x + q\delta y$$

- the problem is
  - p and q are not known
  - image irradiance equation provides only one constraint
  - if we know p and q at given point (x, y) but we need new values of p and q at point $(x+\delta x, y+\delta y)$

$$\delta p = r\delta x + s\delta y \text{ and } \delta q = s\delta x + t\delta y$$

$$\begin{pmatrix} \delta p \\ \delta q \end{pmatrix} = \mathbf{H} \begin{pmatrix} \delta x \\ \delta y \end{pmatrix}$$

$\mathbf{H}$ is Hessian matrix of second partial derivatives

$$\mathbf{H} = \begin{pmatrix} r & s \\ s & t \end{pmatrix}$$



Recovering Depth

- Hessian provides information on the curvature of the surface
- For small surface inclinations, its determinant is the Gaussian curvature
- trace of the Hessian is the Laplacian of depth; for small surface inclinations is twice the so-called mean curvature
- to use Hessian matrix for computing δp and δq; need to know the second partial derivatives of z
- to keep track of them we need still higher derivatives
- Differentiating image irradiance equation with respect to x and y using the chain rule

$$E_x = rR_p + sR_q \text{ and } E_y = sR_q + tR_q$$

$$\begin{pmatrix} E_x \\ E_y \end{pmatrix} = \mathbf{H} \begin{pmatrix} R_p \\ R_q \end{pmatrix}$$

- to solve $\mathbf{H}$ (three unknowns r, s, and t) we have two equations



Recovering Depth

- while we can not continue the solution in an arbitrary direction, we can do so in a specially chosen direction

$$\begin{pmatrix} \delta x \\ \delta y \end{pmatrix} = \begin{pmatrix} R_p \\ R_q \end{pmatrix} \delta\xi$$

$$\begin{pmatrix} \delta p \\ \delta q \end{pmatrix} = \mathbf{H} \begin{pmatrix} \delta x \\ \delta y \end{pmatrix} = \mathbf{H} \begin{pmatrix} R_p \\ R_q \end{pmatrix} \delta\xi = \begin{pmatrix} E_x \\ E_y \end{pmatrix} \delta\xi$$

- if the direction of the change in the image plane is parallel to the gradient of the reflectance map, then the change in (p, q) can be computed
- the direction of the change in gradient space is parallel to the gradient in the image

$$\dot{x} = R_p \quad \dot{y} = R_q \quad \dot{z} = pR_p + qR_q \quad \dot{p} = E_x \quad \dot{q} = E_y$$

dots denote differentiation with respect to ξ

- A solution of these differential equations is a curve on the surface; parameter ξ will vary along this curve



Recovering Depth

- curve traced out by the solutions of the five differential equations are called characteristic curves
- their projections in the image are called base characteristics
- solutions for x, y, z, p and q form a characteristic strip and define surface orientation along this curve
- to obtain the whole surface patch together characteristic strips
- to start solution, it needs a point where initial values are given
- given an initial curve on the surface, a solution for the surface can be obtained as long as this curve is nowhere parallel to any of the characteristics
- on this curve starting values of p and q can be obtained using image irradiance equation $E(x, y) = R(p, q)$ and known derivatives of z along the curve
- i.e. initial curve given in terms of a parameter η, $x(\eta)$, $y(\eta)$ and $z(\eta)$ then along the curve

$$\frac{\partial z}{\partial \eta} = p\frac{\partial x}{\partial \eta} + q\frac{\partial y}{\partial \eta}$$



Singular Points

- Normally an initial curve along with the image of an object not given
- any points where surface orientation can be determined directly?
- Say $R(p, q)$ has a unique isolated maximum at (p_0, q_0)

$$R(p, q) < R(p_0, q_0) \quad \forall (p, q) \neq (p_0, q_0)$$

- at some point (x_0, y_0) in the image $E(x_0, y_0) = R(p_0, q_0)$
- at this point the gradient (p, q) is uniquely determined to be (p_0, q_0)
- we could start the solution at such a singular point, but at a maximum of $R(p, q)$ the R_p and R_q are zero
- solution will not move from such a point because $\dot{x}$ and $\dot{y}$ are zero
- bypass this by constructing a small "cap" at this point and start the solution at the edge of this cap



Shape from Shading: Example

- consider power series near a singular point

$$R(p,q) = \frac{1}{2}(p^2 + q^2)$$

- it has unique isolated minimum, singular point at the origin such that $E(0,0) = 0$, so $(p,q) = (0,0)$ at this point
- say surface is smooth enough, z as a power series (without first order terms) and ignoring higher-order terms near the origin

$$z = z_0 + \frac{1}{2}(ax^2 + 2bxy + cy^2) \quad p = ax + by \quad \text{and} \quad q = bx + cy$$

$$E(x,y) = \frac{1}{2}(a^2 + b^2)x^2 + (a+c)bxy + \frac{1}{2}(b^2 + c^2)y^2$$



Shape from Shading: Example

- task is to determine a, b, and c given the image brightness and its derivatives near the origin

$$z = z_0 - \frac{1}{2}(ax^2 + bxy + \frac{1}{2}cy^2) \quad p = ax + by \quad \text{and} \quad q = bx + cy$$

- gives rise to the same shading pattern; so at least two solutions possible

$$E_x = (a^2 + b^2)x + (a+c)by \quad E_y = (a+c)bx + (b^2 + c^2)y$$

- $(E_x, E_y)^T = (0,0)^T$ at $(x,y) = (0,0)$; we can not use brightness gradient to recover the shape

$$E_{xx} = a^2 + b^2 \quad E_{xy} = (a+c)b \quad E_{yy} = b^2 + c^2$$

- three second order polynomials in three unknowns can have up to eight solutions
- more than one shape might give rise to the same shading, since the nonlinear equations containing the coefficients of the power series near the singular point can have more than one solution



Relaxation Method

- method of characteristic strip expansion suffers from
  - sensitive to measurement noise
  - prevent adjacent characteristics from crossing over each other
- Minimization in continuous case
  - objective is to find two functions $f(x,y)$ and $g(x,y)$ that ensure that the image irradiance equation,

$$E(x,y) = R_s(f,g)$$

    is satisfied. $R_s(f,g)$ reflectance map expressed in stereographic coordinates
  - what we want? $f(x,y)$ and $g(x,y)$ to correspond to a smooth surface
  - to measure smoothness, choose function that penalizes rapid changes in f and g; try to minimize

$$e_s = \int\int_I ((f_x^2 + f_y^2) + (g_x^2 + g_y^2))dxdy$$



Relaxation Method: Solution

- minimize e_s subject to the constraint that f and g must satisfy the image irradiance equation
- in practice, there are errors in both the measurements of irradiance and the determination of reflectance map
- try to minimize the error

$$e_i = \int\int_I (E(x,y) - R_s(f,g))^2 dxdy$$

- overall, minimize $e_s + \lambda e_i$ λ weights the errors in the image irradiance equation relative to the departure from smoothness
- λ should be made large if brightness measurements are very accurate and small if they are very noisy



Relaxation Method: Solution

- minimization of an integral of the form

$$\int\int F(f,g,f_x,f_y,g_x,g_y)dxdy$$

- the corresponding Euler equations are

$$F_f - \frac{\partial}{\partial x}F_{f_x} - \frac{\partial}{\partial y}F_{f_y} = 0 \quad F_g - \frac{\partial}{\partial x}F_{g_x} - \frac{\partial}{\partial y}F_{g_y} = 0$$

$$F = (f_x^2 + f_y^2) + (g_x^2 + g_y^2) + \lambda(E(x,y) - R_s(f,g))^2$$

- aim is to minimize the integral of F and Euler equations for this

$$\nabla^2 f = -\lambda(E(x,y) - R_s(f,g))\frac{\partial R_s}{\partial f} \quad \nabla^2 g = -\lambda(E(x,y) - R_s(f,g))\frac{\partial R_s}{\partial g}$$

where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$

- result is a couple pair of elliptic second-order partial differential equations and can be solved by iterative methods



Minimization in the discrete case

- measure the departure from smoothness at the point (i,j) by

$$s_{i,j} = \frac{1}{4}((f_{i+1,j} - f_{i,j})^2 + (f_{i,j+1} - f_{i,j})^2 + ((g_{i+1,j} - gi,j)^2 + (g_{i,j+1} - g_{i,j}^2))$$

- error in the image irradiance equation

$$r_{ij} = (E_{ij} - R_s(f_{ij}, g_{ij}))^2$$

E_{ij} is the observed image irradiance

- seek a set of values $\{f_{ij}\}$ and $\{g_{ij}\}$ that minimize

$$e = \sum_i \sum_j (s_{ij} + \lambda r_{ij})$$



Minimization in the discrete case

- differentiating e with respect to f_{kl} and g_{kl}

$$\frac{\partial e}{\partial f_{kl}} = 2(f_{kl} - \bar{f}_{kl}) - 2\lambda(E_{kl} - R_s(f_{kl}, g_{kl}))\frac{\partial R_s}{\partial f}$$

$$\frac{\partial e}{\partial g_{kl}} = 2(g_{kl} - \bar{g}_{kl}) - 2\lambda(E_{kl} - R_s(f_{kl}, g_{kl}))\frac{\partial R_s}{\partial g}$$

$$\bar{f}_{i,j} = \frac{1}{4}(f_{i+1,j} + f_{i,j+1} + f_{i-1,j} + f_{i,j-1}) \quad \bar{g}_{i,j} = \frac{1}{4}(g_{i+1,j} + g_{i,j+1} + g_{i-1,j} + g_{i,j-1}$$

- the extremum is to be found where the above derivatives of e are equal to zero
- rearranging the resulting equations by solving for f_{kl} and g_{kl}, an iterative solution method

$$f_{kl}^{n+1} = \bar{f}_{kl}^n + \lambda(E_{kl} - R_s(f_{kl}^n, g_{kl}^n))\frac{\partial R_s}{\partial f} \quad g_{kl}^{n+1} = \bar{g}_{kl}^n + \lambda(E_{kl} - R_s(f_{kl}^n, g_{kl}^n))\frac{\partial R_s}{\partial g}$$



Minimization in the discrete case

if there are n images

$$e = \int\int_I ((f_x^2 + f_y^2) + (g_x^2 + g_y^2))dxdy + \sum_{i=1}^n \lambda_i \int\int_I (E_i(x,y) - R_i(f,g))^2 dxdy$$

$$\nabla^2 f = -\sum_{i=1}^n \lambda_i (E_i(x,y) - R_i(f,g))\frac{\partial R_i}{\partial f}$$

$$\nabla^2 g = -\sum_{i=1}^n \lambda_i (E_i(x,y) - R_i(f,g))\frac{\partial R_i}{\partial g}$$

$$f_{kl}^{n+1} = \bar{f}_{kl}^n + \sum_{i=1}^n \lambda_i (E_{i;kl} - R_i(f_{kl}, g_{kl}))\frac{\partial R_i}{\partial f}$$

$$g_{kl}^{n+1} = \bar{g}_{kl}^n + \sum_{i=1}^n \lambda_i (E_{i;kl} - R_i(f_{kl}, g_{kl}))\frac{\partial R_i}{\partial g}$$



Recovering depth

- given p and q, the partial derivatives of $z(x, y)$ with respect to x and y; recover $z(x, y)$ by integrating along arbitrary curves in the plane

$$z(x, y) = z(x_0, y_0) + \int_{(x_0, y_0)}^{(x,y)} (pdx + qdy)$$

- p and q are recovered from noisy image data by some method
- choose $z(x, y)$ so as to minimize the error

$$\int \int_I ((z_x - p)^2 + (z_y - q)^2) dxdy$$

p and q are the given estimates of the components of the gradient and z_x and z_y are the partial derivatives of the best-fit surface

- this is a problem in the calculus of variations



Recovering depth

- minimize integral of the form

$$\int \int F(z, z_x, z_y) dxdy$$

- Euler equation

$$F_z - \frac{\partial}{\partial x} F_{z_x} - \frac{\partial}{\partial y} F_{z_y} = 0$$

from

$$F = (z_x - p)^2 + (z_y - q)^2$$

we obtain

$$\frac{\partial}{\partial x}(z_x - p) + \frac{\partial}{\partial y}(z_y - q) = 0 \quad \text{or} \quad \nabla^2 z = p_x + q_y$$

